

ASAD ULLAH DOGAR

AI Developer & Backend Engineer

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PROFESSIONAL SUMMARY

AI Developer and Backend Engineer with a **Bachelor of Science in Artificial Intelligence** (DUET) and proven, hands-on experience architecting **end-to-end ML pipelines**, **production-grade asynchronous backend systems**, and **computer vision applications**. Proficient in **Python**, **PyTorch**, and **FastAPI**, equally fluent in designing transfer-learning CV pipelines and hardening microservice security to a Zero-Trust standard. Targeting entry-level **AI Developer** and **Backend Engineer** roles where rigorous engineering and applied research intersect.

EDUCATION

Bachelor of Science in Artificial Intelligence 2026

Dawood University of Engineering and Technology (DUET), Karachi

Key Focus Areas: Computer Vision, Large Language Models (LLMs), Knowledge Distillation

KEY PROJECTS

Custom Teachable Machine - Full-Stack AI System

PyTorch | MobileNetV3 | Scikit-Learn | FastAPI | Streamlit | Transfer Learning

- **Architected** a production-oriented, decoupled client-server ML system using **Streamlit** (UI) and **FastAPI** (backend), reducing cross-layer coupling and enabling independent development, testing, and deployment of each service.
- **Engineered** a dynamic **computer vision pipeline** that ingests live image inputs over a web network, trains a custom classification model on-the-fly, and delivers real-time predictions — compressing the full ML iteration cycle to under 60 seconds end-to-end.
- **Accelerated** feature extraction throughput by integrating a pre-trained **PyTorch MobileNetV3** backbone with a **Scikit-Learn** logistic regression head, achieving high-accuracy localized training through a **transfer-learning** workflow directly portable to RL observation encoders.
- **Designed** the inference service for horizontal scalability by fully decoupling the training loop from the prediction backend, mirroring modular architecture patterns used in production research systems.

Advanced Asynchronous Backend System - Enterprise-Grade Microservice

FastAPI | SQLAlchemy 2.0 | Redis | asyncpg | JWT | GitHub Actions | Docker | Pytest

- **Eliminated** blocking I/O across the entire request lifecycle by architecting a 100% asynchronous **FastAPI** microservice using **SQLAlchemy 2.0**, **aiosqlite/asyncpg**, and **redis-asyncio**, achieving fully **non-blocking I/O** throughput suitable for high-volume data collection pipelines.
- **Hardened** system security to a production-grade **Zero-Trust model** by implementing **PBKDF2-SHA256** password hashing, a dual-token **JWT** system (Access + Refresh tokens), and a **Redis** token blacklist to prevent session-replay attacks.
- **Reduced** integration defect exposure to near-zero by establishing a full **CI/CD pipeline** via **GitHub Actions**, integrating **Pytest** for E2E coverage, **Ruff** for linting, and **Bandit** for static security auditing — enforcing professional codebase quality on every commit.
- **Demonstrated** full-stack system ownership by independently designing the service from database schema to deployment pipeline, delivering a research-grade microservice with zero external architectural guidance.

WORK EXPERIENCE

AI & Data Analytics Intern *Jun 2025 - Aug 2025*

Shaukat Khanum Memorial Cancer Hospital, Karachi

- **Reclaimed** 5+ engineering hours per week by automating the end-to-end manual reporting pipeline through scripted **Python** ETL workflows, eliminating all repetitive data aggregation and formatting tasks.
- **Improved** dashboard responsiveness by **30%** by optimizing query execution plans and refactoring **Plotly** rendering logic, enabling real-time engagement monitoring for senior program stakeholders.
- **Empowered** non-technical staff to self-serve analytics by designing interactive **Streamlit** and **Plotly** dashboards, significantly reducing ad-hoc data requests to the engineering team.
- **Increased** outreach targeting precision by developing a cohort classification model achieving **84% accuracy** using **Scikit-Learn**, enabling data-driven allocation of limited program resources.
- **Preserved** institutional knowledge and long-term infrastructure sustainability by authoring comprehensive handover documentation covering pipeline architecture, data schemas, and dashboard maintenance procedures.

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, Scikit-Learn, Transfer Learning, Computer Vision, LLMs, Knowledge Distillation, Predictive Modeling, Feature Engineering

Backend & APIs: Python, FastAPI, Django, Flask, RESTful APIs, SQLAlchemy 2.0, Async I/O, JWT Auth, Redis, Cache Management

Data & Visualization: Pandas, NumPy, Streamlit, Plotly, Matplotlib, Seaborn, OpenCV

DevOps & Databases: GitHub Actions, Docker, Pytest, PostgreSQL, SQLite, Git

Scraping & Tooling: BeautifulSoup, Selenium

LEADERSHIP & SOCIAL IMPACT

Adaptation Ambassador

Climate Resilience & Community Outreach Initiative, Karachi

- **Spearheaded** community-level climate adaptation programs across the Karachi region, equipping youth cohorts with actionable resilience strategies aligned with national and international sustainability frameworks.
- **Delivered** climate policy literacy workshops that translated complex environmental data into accessible, community-oriented messaging, measurably increasing awareness of adaptation frameworks among underserved populations.

Student Ambassador

Shaukat Khanum Memorial Cancer Hospital

- **Championed** healthcare accessibility awareness drives and fundraising campaigns, directly contributing to the hospital's community outreach mission and patient support infrastructure.

CERTIFICATIONS

- **AWS AI Practitioner Challenge** - Udacity (Mar 2026)
- **AI Fluency for Students** - Anthropic (Sep 2025)
- **Google Prompting Essentials & Introduction to Generative AI** - Google Cloud
- **Introduction to AI Agents, Agentic AI & LLMs** - 365 Data Science
- **AI For Everyone** - Coursera (Andrew Ng)
- **100 Days of Code: Complete Python Pro Bootcamp** - Udemy